

五. 系综理论

1. 基本假设: 1) 粒子总能量 $H\psi = \mathcal{E}\psi$, $H = \sum \frac{p_i^2}{2m} + \sum_{i,j} u_{ij}(r_{ij})$ (暂不考虑势能 $V(r)$)
($\Omega \neq 0$) 2) 状态数: $\mathcal{E} \in [E, E+\Delta E]$, $p \in [p, p+\Delta p]_{dim}$ $\rightarrow \Omega = \Omega(E, \Delta E; p, \Delta p \dots)$
L 平均概率假设: 平衡态, 上述每个态等概, $P = \frac{1}{\Omega}$ (较普遍的基本假设)

1) 多次重复等概 2) 一次多个等概 (掷骰子与掷多个骰子的定义等价) ? 条件

2. 正则系综 (N, E 不变)

N 个低速量子: $H\psi_j = E_j\psi_j$, $H = \sum \frac{p_i^2}{2m} + \sum_{i,j} u_{ij}(r_{ij})$

系综: M 个相同系统组成系综, 系统间有能量交换, 也与外界交换:

$$H_{\text{系综}} = \sum_{\alpha=1}^M H_{\alpha} + \Delta E \text{ (热交换项)}$$

正则系综: 与外界热交换使温度一定 $\rightarrow$ 系统较大时可忽略热交换项



条件: 1) $H_{\text{系综}} = \sum_{\alpha=1}^M H_{\alpha}$

2) $\Psi_{\text{系综}} = \prod_{\alpha=1}^M \psi_{\alpha}$

相关量: $M = \sum_j M_j \rightarrow \psi_j$ 上的系统数

$$\mathcal{E} = \sum_j M_j E_j$$

$$\Omega = \frac{M!}{\prod_j M_j!} \text{ (同-态系统可交换) } \Omega \text{ 为 } [M_j] \text{ 的态数, } P_{[M_j]} = \frac{\Omega}{\text{all } \Omega} \rightarrow \max \Omega \text{ 为最概然系综}$$

$$\delta \ln \Omega = \frac{\partial \ln \Omega}{\partial M_j} \cdot \delta M_j \xrightarrow{\text{拉格朗日}} \frac{\partial \ln \Omega}{\partial M_j} - \alpha \frac{\partial \sum M_j}{\partial M_j} - \beta \frac{\partial \sum M_j E_j}{\partial M_j} = 0$$

$$\ln \Omega = M \ln M - M - \sum M_j \ln M_j + \sum M_j \Rightarrow \text{原式: } -\ln M_j - \alpha \cdot 1 - \beta \cdot E_j = 0 \rightarrow M_j = e^{\alpha - \beta E_j}$$

最概然分布的系综: $P_j \equiv \frac{M_j}{M} = \frac{e^{-\beta E_j}}{\sum_i e^{-\beta E_i}} \rightarrow$ 配分函数 $Q \equiv \sum_i e^{-\beta E_i}$ (作为归一化因子出现)

平均能 $E \equiv \frac{\mathcal{E}}{M} = \frac{1}{Q} \sum E_j e^{-\beta E_j} \rightarrow E$ 给定, $M \rightarrow \infty$ 时, P_i 与 β, M 无关

$$f(M_j) = \ln \Omega - \alpha \sum M_j - \beta \sum M_j E_j, \quad \frac{\partial f}{\partial M_j} = 0 \text{ 取极值, } M_j = MP_j$$
$$\frac{\partial^2 f}{\partial m_i^2} = -\frac{1}{m_i} < 0, \text{ 展开: } f(m_i) = f(\bar{m}_i) + \underbrace{\sum \frac{\partial f}{\partial m_i} (m_i - \bar{m}_i)}_0 + \frac{1}{2} \sum \frac{\partial^2 f}{\partial m_i^2} (m_i - \bar{m}_i)^2 + \dots$$

$$\text{误差 } \sigma_j = \sqrt{\frac{M_j^2 - \bar{M}_j^2}{\bar{M}_j^2}} = \sqrt{\frac{MP_j}{MP_j^2}} \rightarrow 0$$

$$f(M_j)_{\text{max}} = \ln \bar{\Sigma} - \alpha M - \beta M E$$

物理量: $\beta \equiv \frac{1}{kT}$ $F = -kT \ln Q$ $\xrightarrow{\text{熵自由能}}$ $S = -\frac{\partial F}{\partial T}$, $P = \frac{\partial F}{\partial V}$, $f(P, V, T) = 0$

Pf: $dF_{\text{ph}} = -SdT - pdV$, $-kT^2 \frac{\partial}{\partial T} \left(\frac{F_{\text{ph}}}{kT} \right)_{V,N} = -kT^2 \left[\frac{F_{\text{ph}}}{-kT^2} + \frac{S}{kT} \right] = F_{\text{ph}} + TS = E$

$$\frac{\partial \ln \Omega}{\partial T} = \frac{E}{kT} \rightarrow \frac{\partial}{\partial T} \left(\frac{F}{kT} \right) = E \quad \xrightarrow{\quad} \quad \frac{\partial}{\partial T} \left(\frac{F - F_A}{kT} \right) = 0$$

$T \rightarrow 0$ 基态: $Q = w_0 e^{\frac{E_0}{kT}}$, $F = E_0 - kT \ln w_0$ (w_0 简并度)

$$F_{\text{热}} = E_0 - TS_0 \Rightarrow S_0 \equiv k \ln \omega_0 \rightarrow F = F_{\text{热}}$$

定义0标

3. 巨正则系综：交换能量与粒子数 eg: 水与空气: $dE = -pdv + Tds + \mu dN$, $\mu = \frac{G}{N}$



$$H_{\text{系}} = \sum_M H + \text{"相互作用"}$$

交换粒子与能量的贡献

系统较大时, 相互作用忽略 ($H \gg \text{相}$)

- 1) $H_{\text{系}} = \sum H_{\text{块}}$
- 2) $\Psi_{\text{系}} = \prod \psi_{\text{块}}$
- 3) $H(N) |j(N)\rangle = E_{j(N)} |j(N)\rangle$

$M_{j(N)}$ 为 N 粒子在 $|j(N)\rangle$ 态的系数

$$\sum_{N, j(N)} M_{j(N)} = M \quad (\text{总数}), \quad \sum_{N, j(N)} M_{j(N)} \cdot N \equiv M \cdot \mathcal{N} \quad (\text{平均粒子数}), \quad \sum_{N, j(N)} M_{j(N)} \cdot E_{j(N)} \equiv ME$$

方法同上: $\Omega = \frac{M!}{\prod_{N, j(N)} M_{j(N)}!}$, $\ln \Omega \approx M \ln M - M - \sum M_{j(N)} \ln M_{j(N)} + \sum M_{j(N)}$ $\rightarrow \frac{\ln \Omega}{M} = \ln M - \frac{\sum M_{j(N)} \ln M_{j(N)}}{M} = \sum \frac{M_{j(N)}}{M} \cdot \ln \frac{M_{j(N)}}{M} = -\sum P_{j(N)} \ln P_{j(N)}$

$= \ln \Xi + \beta E + \gamma \mathcal{N}$

约束: $M, \mathcal{N}, E \rightarrow \frac{\partial}{\partial M_{j(N)}} [\ln \Omega - \sum M_{j(N)} (\alpha + \beta E_{j(N)} + \gamma N)] = 0$

同上, $M \rightarrow \infty, \beta \sim \frac{1}{M} \rightarrow 0$

巨配分函数: $\Xi = \sum_{N, j(N)} e^{-\beta E_{j(N)} - \gamma N}$, $P_j = \frac{1}{\Xi} \cdot e^{-\beta E_j - \gamma N}$

$$\begin{cases} -(\frac{\partial}{\partial \beta} \ln \Xi)_{\gamma, \gamma} = \frac{\sum E_{j(N)} \cdot e^{-\beta E_j - \gamma N}}{\Xi} = \langle E \rangle = E \\ -(\frac{\partial}{\partial \gamma} \ln \Xi)_{\beta, \beta} = \frac{\sum N \cdot e^{-\beta E_j - \gamma N}}{\Xi} = \langle N \rangle = \mathcal{N} \end{cases}$$

物理量 β : 为与 - 黑体辐射源, 与系统能量交换平衡时 $\beta = \frac{1}{kT}$

γ : $d(\frac{\ln \Omega}{M})_{\gamma} = \underbrace{d(\ln \Xi)_{\gamma}}_{-E d\beta - \mathcal{N} d\gamma} + \beta dE + E d\beta + \gamma d\mathcal{N} + \mathcal{N} d\gamma = \beta dE + \gamma d\mathcal{N} \rightarrow d(\frac{S}{k})_{\gamma, \gamma} = \frac{dE}{kT}, \quad d(S - k \frac{\ln \Omega}{M})_{\gamma, \gamma} = 0 \rightarrow S - k \frac{\ln \Omega}{M} = f(\gamma, \gamma)$

$T \rightarrow 0, \Omega \approx \omega_0^M, \quad kM^{-1} \ln \Omega_0 = k \ln \omega_0$; $S_{\text{热}} = k \ln \omega_0$ $\xrightarrow{\text{定标}}$ $S = k \frac{\ln \Omega}{M} \rightarrow \frac{dS}{k} = \beta dE + \gamma d\mathcal{N} \Rightarrow \gamma = \frac{-\mu}{kT}, \quad \mu = \frac{G}{N}$

$$p_{\text{热}} \equiv \frac{kT}{V} \ln \Xi, \quad p_{\text{力}} \equiv \left\langle - \frac{\partial E_{j(n)}}{\partial V} \right\rangle, \quad V \rightarrow \infty \text{ 时, 等价 } \Rightarrow \frac{p}{kT} = \frac{\ln \Xi}{V}$$

逸度: $z \equiv e^{-\beta \mu} = e^{\frac{\mu}{kT}}, \quad \Xi = \sum z^N \cdot e^{-\frac{E_{j(n)}}{kT}}$

$$N = \sum P_{j(n)}, \quad N = \frac{1}{\Xi} \cdot \sum N z^n e^{-\frac{E_{j(n)}}{kT}} = \frac{\partial \ln \Xi}{\partial \ln z}$$

3. 微正则系综

N, V, E 不变 E 可在 $E \sim E + \Delta E$ 间, $\Delta E \ll E$

1) 总系统 = $\sum_{\alpha=1}^M H_{\alpha}$

$\sum M_{\alpha} = M, \quad \sum M_{\alpha} E_{\alpha} = \mathcal{E} = M \cdot E, \quad \Omega = \frac{M!}{\prod M_j!}$

2) $\Psi_{\text{系统}} = \prod_{\alpha=1}^M \psi_{\alpha}$

$\frac{\partial \ln \Omega}{\partial M_j} \Rightarrow \ln M_j - \alpha - \beta E_j = 0 \rightarrow M_j = e^{-\alpha - \beta E_j}$

4. 费米子, 玻色子

动量 $\vec{p} = \hbar \vec{k}$, 螺旋性 $\lambda = -j \dots j$

$\epsilon_{\alpha} = c \sqrt{\hbar^2 k^2 + m^2 c^2} - mc^2, \quad E = \sum_{\alpha} n_{\alpha} \epsilon_{\alpha}$

费: $n_{\alpha} = 0, 1$
玻: n_{α}

巨配分函数 $\mathcal{Q} = \sum z^{\sum n_{\alpha}} e^{-\frac{\sum n_{\alpha} \epsilon_{\alpha}}{kT}} = \prod_{\alpha} \sum_{n_{\alpha}} z^{n_{\alpha}} e^{-\frac{n_{\alpha} \epsilon_{\alpha}}{kT}}$

费: $\mathcal{Q}_{\epsilon=0} = \prod (1 + z e^{-\frac{\epsilon_{\alpha}}{kT}})$

玻: $\mathcal{Q}_{\epsilon=0} = \prod \frac{1}{(1 - z e^{-\frac{\epsilon_{\alpha}}{kT}})}$

$p_{\text{热}} = \frac{kT}{V} \ln \mathcal{Q} \rightarrow \frac{pV}{kT} = \ln \mathcal{Q} = \sum_{\alpha} \pm \ln(1 \pm z e^{-\frac{\epsilon_{\alpha}}{kT}})$ { 费: +
玻: - }

$\sum_{\alpha} \rightarrow \frac{\omega}{2\pi^3} \int \frac{d^3 k}{2\pi^3} \frac{1}{z^{\pm} e^{\frac{\epsilon_{\alpha}}{kT}} \pm 1}$

平均数 $\langle n_{\alpha} \rangle = \frac{\sum n_{\alpha} p_{\alpha}}{\sum p_{\alpha}}$ { 费米: $\langle n_{\alpha} \rangle = \frac{0 + 1 \cdot z e^{-\frac{\epsilon_{\alpha}}{kT}}}{1 + z e^{-\frac{\epsilon_{\alpha}}{kT}}} = \frac{z e^{-\frac{\epsilon_{\alpha}}{kT}}}{1 + z e^{-\frac{\epsilon_{\alpha}}{kT}}}$
玻色: $\langle n_{\alpha} \rangle = \frac{0 \cdot 1 + 1 \cdot z + 2 \cdot z^2 + \dots}{1 + z + z^2 + \dots} = x \frac{d}{dx} \ln \frac{1}{1-x} = \frac{z e^{-\frac{\epsilon_{\alpha}}{kT}}}{1 - z e^{-\frac{\epsilon_{\alpha}}{kT}}}$

$\frac{pV}{kT} = \frac{\omega}{2\pi^3} \int d^3 k \ln(1 \pm z e^{-\frac{\epsilon_{\alpha}}{kT}})$ { 费: +
玻: - }

$\langle n \rangle = \frac{1}{e^{\frac{\epsilon_{\alpha}}{kT}} \pm 1}$

熵散: $\frac{p}{kT} = \frac{\omega}{2\pi^3} \cdot \frac{1}{V} \int d^3 k \ln(1 \pm z e^{-\frac{\epsilon_{\alpha}}{kT}})$
= ?

低速: $\epsilon = \frac{\hbar^2 k^2}{2m} \rightarrow d^3 k \propto \epsilon^2 d\epsilon$, 高速 $\epsilon = \hbar c |k| \rightarrow d^3 k \propto \epsilon^2 d\epsilon$

$\pm \int \epsilon^2 d\epsilon \ln(1 \pm z e^{-\frac{\epsilon_{\alpha}}{kT}}) = \pm \frac{\epsilon^3}{3} \ln(1 \pm z e^{-\frac{\epsilon_{\alpha}}{kT}}) \Big|_0^{\infty} + \int \frac{\epsilon^3}{3} \frac{z e^{-\frac{\epsilon_{\alpha}}{kT}}}{1 \pm z e^{-\frac{\epsilon_{\alpha}}{kT}}} \frac{d\epsilon}{kT} \rightarrow$ 对 $\epsilon^2 d\epsilon \omega d^3 k$, 经典: $n = \frac{1}{2}$, 高速: $n = 2$

$\frac{\epsilon^3 d\epsilon}{n+1} \leftrightarrow \frac{\epsilon d^3 k}{n+1}$

$N = \sum \langle n_{\alpha} \rangle = \frac{\omega}{2\pi^3} V \int d^3 k \langle n_{\alpha} \rangle \rightarrow E = \frac{\omega}{2\pi^3} V \int d^3 k \epsilon \langle n_{\alpha} \rangle \rightarrow \frac{pV}{kT} = \frac{1}{n+1} \frac{1}{kT} \cdot \frac{\omega V}{2\pi^3} \int d^3 k \cdot \epsilon \langle n_{\alpha} \rangle \Rightarrow pV = \frac{E}{n+1}$

经典: $pV = \frac{2}{3} E$

相对论: $pV = \frac{1}{3} E$

5. 非理想气体:

基本公式: $\frac{P}{kT} = \frac{1}{V} \ln Q$

$Q = \sum_{y=0}^{\infty} \frac{y^N}{N!} Q_N(N)$ (对粒子数展开)

$Q_N(N) = \int e^{\frac{-U_N}{kT}} \prod_{i=1}^N d^3r_i$

$y = \frac{q}{\lambda^3} z = \frac{q}{\sqrt{\frac{2\pi}{mkT}}} h e^{\frac{\mu}{kT}}$

对自由粒子: $U_{ij}=0 \Rightarrow f_{ij}=0 \Rightarrow b_1=1, b_1=0 \Rightarrow \frac{P}{kT}=y, P=y$

$\frac{P}{kT}=y, \text{理想气体}$

对相互作用: $U_{ij} \neq 0 \Rightarrow \frac{P}{kT} = y + b_2 y^2 + \dots, \rho = y + 2b_2 y^2 + \dots$

故 $\frac{P}{kT} = \rho - b_2 \rho^2 + O(\rho^3)$

$y = \rho - 2b_2 \rho^2 + \dots$

位力展开, $b_2(V) = \frac{1}{2!V} \int f_{12} d^3r_1 d^3r_2$, 取刚球模型 $U_{ij} = \begin{cases} 0 & r > d \\ -\frac{u_0}{kT} & r \leq d \end{cases}$ (展开后)

刚球 $b_2(\infty) = -\frac{1}{2} \frac{u_0}{kT} d^3 = -\frac{1}{2kT} \int u(r) d^3r$

范氏: $P^* V^* = NkT \Rightarrow \frac{P}{kT} = \frac{P^*}{P_b^*} - \frac{P^* a^*}{kT} \cdot \frac{1}{V^*}$, 比较得: $b_2 = \frac{a}{kT} - b \Rightarrow a = -\frac{1}{2} \int u d^3r = -\frac{1}{2} \sqrt{2} u_0$

三阶: $\frac{P}{kT} = y + b_2 y^2 + b_3 y^3 = \rho - b_2 \rho^2 + (4b_2^2 - 2b_3) \rho^3 + O(\rho^3)$

$b_3 = \frac{1}{6V} \int [f_{12} f_{13} + f_{12} f_{23} + f_{13} f_{23}] d^3r_1 d^3r_2 d^3r_3 = \frac{1}{6V} \int f_{12} f_{13} d^3r_{12} + \frac{1}{6V} \int f_{12} f_{23} d^3r_{12}$

$\downarrow$
 $4b_2^2 - 2b_3 = -\frac{1}{3V} \int f_{12} f_{23} d^3r_1 d^3r_2 d^3r_3 = \frac{1}{2} (2b_2)^2$ (恰消去)

粒子势能 $U_N = \sum_{i,j} U_{ij}$ (要求: $\frac{1}{r}$ 衰减 (库仑能可正负抵消))

对因子 $e^{\frac{-U_{ij}}{kT}} \rightarrow r \rightarrow \infty$ 时趋 1 $\Rightarrow f_{ij} = e^{\frac{-U_{ij}}{kT}} - 1 \xrightarrow{r \rightarrow \infty} 0, f_{ij} = f(r_{ij})$

$e^{\frac{-U}{kT}} = e^{\sum_{i,j} \frac{-U_{ij}}{kT}} = \prod (1 + f_{ij}) = 1 + \sum f_{ij} + \sum f_{ij} f_{ij} + \dots$ (展开)

	1粒子	2粒子	n粒子
图示:			
值:	1	1 + f_{12}	1 + \sum f_{ij} + \dots

集团积分: $b_L(T, V), b_1 \equiv \frac{1}{V} \int \prod_{i=1}^L d^3r_i \sum_{L \in \text{集团}} (\prod f_{ij})$

性质: 1) b_L 无量纲 (f_{ij} 无量纲, d^3r 与 V 抵) $\star$ 不为比

2) $V \rightarrow \infty$ 时为 $b_L(T)$: $V \rightarrow \infty$ 时 d^3r_i 积分, $V \rightarrow \infty$ 使 f_{ij} 积分不受其他粒子影响

$b_1 = \frac{1}{V} [0] = \frac{1}{V} \int d^3r = 1$

$b_2 = \frac{1}{V^2} [0-1] = \frac{1}{2V} \int f_{12} d^3r_1 d^3r_2 \dots$

迈耶第二定理: $(\frac{P}{kT})_V = \sum_{L=1}^{\infty} b_L(V, T) y^L, \rho = \sum_{L=1}^{\infty} L b_L y^L$

pf: $Q = \sum_{N!} y^N Q_N(N)$, N 个点分 n_L 个群, 粒数 $\leq N$ 的分法有 $\frac{N!}{\prod (L!)^{n_L} n_L!}$

对 $e^{\frac{-U}{kT}}$ 展成 f_{ij} 级, 对相同因子相加积分: $\prod_L (b_L V \cdot L!)^{n_L}$

$\frac{N!}{\prod (L!)^{n_L} n_L!} \prod_L (b_L V \cdot L!)^{n_L} = N! \prod_L \frac{1}{n_L!} (b_L V)^{n_L}$

对 $Q_N(N) = \sum_{\{n_L\}} N! \prod_L \frac{1}{n_L!} (b_L V)^{n_L}$

$\downarrow$
 $Q = \sum_{N!} y^N Q_N(N) = \sum_{\{n_L\}} \prod_L \frac{y^{n_L L}}{n_L!} (b_L V)^{n_L} = \prod_{L=1}^{\infty} \sum_{n_L=0}^{\infty} \frac{1}{n_L!} (b_L V y^L)^{n_L} = e^{\sum_{L=1}^{\infty} b_L V y^L}$

N 是最大, 故 n_L 无限制

故 $\frac{P}{kT} = \sum_{L=1}^{\infty} b_L y^L, \rho = \frac{\partial}{\partial y} (\frac{P}{kT}) = \sum_{L=1}^{\infty} L b_L y^L$

不可约集团: 去除粒子线不使之成2团 (含 $o-o$)

$$\text{不可约集团积分: } \beta_k = \frac{1}{k!} \frac{1}{V} \int \sum_{\text{不可约 } k+1 \text{ 团}} \prod_{i=1}^k d\vec{r}_i \prod_{ij} f_{ij}$$

$$| \beta_2 = \frac{1}{V} \int f_{12} d\vec{r}_1 d\vec{r}_2 \quad o-o$$

$$\beta_3 = \frac{1}{2V} \int f_{12} f_{13} f_{23} d\vec{r}_{123} \quad \text{triangle}, \quad b_3 = \frac{1}{2} \beta_1^3 + \frac{1}{3} \beta_2$$

$$\text{迈耶第二定理: } \frac{P}{kT} = \rho \left[1 - \sum_{k=2}^{\infty} \frac{\beta_k}{k} \rho^k \right]$$

待证(见后)